

NASA C-MAPSS FD003 — Parallel CNN–LSTM Reconstruction

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Task: Remaining-useful-life prediction

Architecture: Parallel CNN and LSTM branches with feature fusion

This notebook reconstructs the parallel FD003 experiment in the style of the MSc thesis notebooks while correcting reproducibility problems found in the historical files.

Result status: The thesis reports a test RMSE of **14.75** for the parallel model. That number is a historical reference, not a saved output of this notebook. Run every cell with the authorized FD003 files before reporting a reproduced result.

Reproduction rules used here

- The 100 training engines are split at engine level into training and validation groups.
- Standardization is fitted on all training engines before the train/validation split.
- The official NASA test trajectories and RUL labels are evaluated once, after training decisions.
- The original sequence length, RUL cap, sensor subset, smoothing factor, and parallel architecture are retained.
- Paths are project-relative and all generated artifacts are written outside the notebook folder.

Environment

Use Python 3.10 or 3.11 and install the pinned project requirements before running. Raw NASA data is not distributed with this repository.

```
from __future__ import annotations

import json
import os
import platform
import random
import sys
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
import sklearn
import tensorflow as tf
from sklearn.metrics import mean_absolute_error, mean_squared_error,
```

```

r2_score
from sklearn.model_selection import GroupShuffleSplit
from sklearn.preprocessing import StandardScaler
from tensorflow.keras import Model, regularizers
from tensorflow.keras.callbacks import EarlyStopping, ModelCheckpoint,
ReduceLROnPlateau
from tensorflow.keras.layers import (
    Concatenate,
    Conv1D,
    Dense,
    Dropout,
    Flatten,
    Input,
    Lambda,
    LSTM,
    Masking,
    MaxPooling1D,
)
from tensorflow.keras.optimizers import Adam

sns.set_theme(style="whitegrid")

SEED = 42

def set_all_seeds(seed: int) -> None:
    """Reset every source of randomness so a training run is
    reproducible for that seed."""
    os.environ["PYTHONHASHSEED"] = str(seed)
    random.seed(seed)
    np.random.seed(seed)
    tf.keras.utils.set_random_seed(seed)

set_all_seeds(SEED)

try:
    tf.config.experimental.enable_op_determinism()
except Exception:
    # Deterministic kernels are not available in every TensorFlow
    build.
    pass

for gpu in tf.config.list_physical_devices("GPU"):
    try:
        tf.config.experimental.set_memory_growth(gpu, True)
    except RuntimeError:
        pass

print("Python:", sys.version.split()[0])

```

```

print("TensorFlow:", tf.__version__)
print("NumPy:", np.__version__)
print("pandas:", pd.__version__)
print("scikit-learn:", sklearn.__version__)
print("GPU devices:", tf.config.list_physical_devices("GPU"))

Python: 3.8.0
TensorFlow: 2.10.0
NumPy: 1.23.5
pandas: 2.0.3
scikit-learn: 1.3.2
GPU devices: [PhysicalDevice(name='/physical_device:GPU:0',
device_type='GPU')]

```

Locate the repository, dataset, and output folders

```

def find_project_root(start: Path | None = None) -> Path:
    """Find the repository root whether the notebook starts there or
    in its own folder."""
    start = (start or Path.cwd()).resolve()
    for candidate in (start, *start.parents):
        if (candidate / "datasets" / "cmapss_fd003").is_dir():
            return candidate
    raise FileNotFoundError(
        "Could not find the predictive-maintenance-rul repository
        root. "
        "Run the notebook from inside the repository or set
        PM_RUL_PROJECT_ROOT."
    )

project_root_override = os.getenv("PM_RUL_PROJECT_ROOT")
PROJECT_ROOT = (
    Path(project_root_override).expanduser().resolve()
    if project_root_override
    else find_project_root()
)

data_dir_override = os.getenv("CMAPSS_FD003_DATA_DIR")
DATA_DIR = (
    Path(data_dir_override).expanduser().resolve()
    if data_dir_override
    else PROJECT_ROOT / "datasets" / "cmapss_fd003" / "data"
)

ARTIFACT_DIR = PROJECT_ROOT / "artifacts" / "cmapss_fd003"
ARTIFACT_DIR.mkdir(parents=True, exist_ok=True)

TRAIN_FILE = DATA_DIR / "train_FD003.txt"
TEST_FILE = DATA_DIR / "test_FD003.txt"

```

```
RUL_FILE = DATA_DIR / "RUL_FD003.txt"
```

```
print("PROJECT_ROOT:", PROJECT_ROOT)
```

```
print("DATA_DIR:", DATA_DIR)
```

```
print("ARTIFACT_DIR:", ARTIFACT_DIR)
```

```
PROJECT_ROOT: C:\Users\Kamal\Documents\AUC\thesiswork\env_eweda\
predictive-maintenance-rul_working
```

```
DATA_DIR: C:\Users\Kamal\Documents\AUC\thesiswork\env_eweda\
predictive-maintenance-rul_working\datasets\cmapss_fd003\data
```

```
ARTIFACT_DIR: C:\Users\Kamal\Documents\AUC\thesiswork\env_eweda\
predictive-maintenance-rul_working\artifacts\cmapss_fd003
```

Load NASA C-MAPSS FD003

```
index_names = ["unit_nr", "time_cycles"]
```

```
setting_names = ["setting_1", "setting_2", "setting_3"]
```

```
sensor_names = [f"s_{i}" for i in range(1, 22)]
```

```
column_names = index_names + setting_names + sensor_names
```

```
def require_files(paths: list[Path]) -> None:
```

```
    missing = [str(path) for path in paths if not path.is_file()]
```

```
    if missing:
```

```
        raise FileNotFoundError(
```

```
            "Missing NASA C-MAPSS FD003 files:\n- "
```

```
            + "\n- ".join(missing)
```

```
            + "\nPlace train_FD003.txt, test_FD003.txt, and
```

```
RUL_FD003.txt "
```

```
            "in datasets/cmapss_fd003/data, or set
```

```
CMAPSS_FD003_DATA_DIR."
```

```
)
```

```
require_files([TRAIN_FILE, TEST_FILE, RUL_FILE])
```

```
train = pd.read_csv(TRAIN_FILE, sep=r"\s+", header=None,
names=column_names)
```

```
test = pd.read_csv(TEST_FILE, sep=r"\s+", header=None,
names=column_names)
```

```
y_test = pd.read_csv(RUL_FILE, sep=r"\s+", header=None, names=["RUL"])
```

```
print("Train shape:", train.shape)
```

```
print("Test trajectory shape:", test.shape)
```

```
print("Official test labels:", y_test.shape)
```

```
display(train.head())
```

```
Train shape: (24720, 26)
```

```
Test trajectory shape: (16596, 26)
```

```
Official test labels: (100, 1)
```

	unit_nr	time_cycles	setting_1	setting_2	setting_3	s_1
s_2 \						
0	1	1	-0.0005	0.0004	100.0	518.67
642.36						
1	1	2	0.0008	-0.0003	100.0	518.67
642.50						
2	1	3	-0.0014	-0.0002	100.0	518.67
642.18						
3	1	4	-0.0020	0.0001	100.0	518.67
642.92						
4	1	5	0.0016	0.0000	100.0	518.67
641.68						

	s_3	s_4	s_5	...	s_12	s_13	s_14	s_15
s_16 s_17 \								
0	1583.23	1396.84	14.62	...	522.31	2388.01	8145.32	8.4246
0.03 391								
1	1584.69	1396.89	14.62	...	522.42	2388.03	8152.85	8.4403
0.03 392								
2	1582.35	1405.61	14.62	...	522.03	2388.00	8150.17	8.3901
0.03 391								
3	1585.61	1392.27	14.62	...	522.49	2388.08	8146.56	8.3878
0.03 392								
4	1588.63	1397.65	14.62	...	522.58	2388.03	8147.80	8.3869
0.03 392								

	s_18	s_19	s_20	s_21
0	2388	100.0	39.11	23.3537
1	2388	100.0	38.99	23.4491
2	2388	100.0	38.85	23.3669
3	2388	100.0	38.96	23.2951
4	2388	100.0	39.14	23.4583

[5 rows x 26 columns]

Construct capped RUL targets

```

RUL_CAP = 125
SEQUENCE_LENGTH = 30
TRAIN_FRACTION = 0.80
MASK_VALUE = -99.0

# SEQUENCE_LENGTH was raised from the original 20: more temporal
# context per
# window is a common choice in the FD003 literature (often 30-50), and
# is
# untested here so far, unlike the regularization/monitoring changes
# above
# that were tried and reverted. Still well under the shortest official

```

```

test
# trajectory (38 cycles), so no test window needs padding.

def add_remaining_useful_life(df: pd.DataFrame, cap: int = RUL_CAP) ->
pd.DataFrame:
    """Create piecewise-linear RUL targets used in the thesis
    experiments."""
    result = df.copy()
    max_cycle = result.groupby("unit_nr")
["time_cycles"].transform("max")
    result["RUL"] = (max_cycle -
result["time_cycles"]).clip(upper=cap)
    return result

train = add_remaining_useful_life(train)

```

Select sensors, fit scalers on all engines, smooth trajectories, and split by engine

```

# The retained 14 sensors match the original C-MAPSS notebook family.
remaining_sensors = [
    "s_2", "s_3", "s_4", "s_7", "s_8", "s_9", "s_11",
    "s_12", "s_13", "s_14", "s_15", "s_17", "s_20", "s_21",
]
drop_sensors = [sensor for sensor in sensor_names if sensor not in
remaining_sensors]

def add_operating_condition(df: pd.DataFrame) -> pd.DataFrame:
    result = df.copy()
    result["setting_1"] = result["setting_1"].round()
    result["setting_2"] = result["setting_2"].round(2)
    result["op_cond"] = (
        result["setting_1"].astype(str)
        + "_" + result["setting_2"].astype(str)
        + "_" + result["setting_3"].astype(str)
    )
    return result

def fit_condition_scalers(
    df: pd.DataFrame,
    sensors: list[str],
) -> tuple[dict[str, StandardScaler], StandardScaler]:
    """Fit scaling on all training engines, before the
    train/validation split."""
    condition_scalers: dict[str, StandardScaler] = {}

```

```

        for condition, group in df.groupby("op_cond"):
            condition_scalers[str(condition)] =
StandardScaler().fit(group[sensors])
        global_scaler = StandardScaler().fit(df[sensors])
        return condition_scalers, global_scaler

def transform_conditions(
    df: pd.DataFrame,
    sensors: list[str],
    condition_scalers: dict[str, StandardScaler],
    global_scaler: StandardScaler,
) -> pd.DataFrame:
    result = df.copy()
    for condition, index in result.groupby("op_cond").groups.items():
        scaler = condition_scalers.get(str(condition), global_scaler)
        result.loc[index, sensors] =
scaler.transform(result.loc[index, sensors])
    return result

def exponential_smoothing(
    df: pd.DataFrame,
    sensors: list[str],
    n_samples: int = 0,
    alpha: float = 0.4,
) -> pd.DataFrame:
    result = df.copy()
    result[sensors] = result.groupby("unit_nr")[sensors].transform(
        lambda values: values.ewm(alpha=alpha).mean()
    )
    if n_samples:
        result = result[result.groupby("unit_nr").cumcount() >=
n_samples].copy()
    return result

train_interim =
add_operating_condition(train.drop(columns=drop_sensors))
test_interim =
add_operating_condition(test.drop(columns=drop_sensors))

condition_scalers, global_scaler = fit_condition_scalers(
    train_interim,
    remaining_sensors,
)

train_scaled = transform_conditions(
    train_interim,
    remaining_sensors,

```

```

        condition_scalers,
        global_scaler,
    )
test_scaled = transform_conditions(
    test_interim,
    remaining_sensors,
    condition_scalers,
    global_scaler,
)

train_scaled = exponential_smoothing(train_scaled, remaining_sensors,
n_samples=0, alpha=0.4)
test_scaled = exponential_smoothing(test_scaled, remaining_sensors,
n_samples=0, alpha=0.4)

print("Retained sensors:", remaining_sensors)
print("Operating conditions fitted from all training engines:",
len(condition_scalers))

# Engine-level split happens after scaling, so it has no effect on the
fitted scalers.
all_train_units = np.sort(train_scaled["unit_nr"].unique())
splitter = GroupShuffleSplit(
    n_splits=1,
    train_size=TRAIN_FRACTION,
    random_state=SEED,
)
train_idx, val_idx = next(splitter.split(all_train_units,
groups=all_train_units))
train_units = all_train_units[train_idx]
val_units = all_train_units[val_idx]

assert set(train_units).isdisjoint(set(val_units))
assert set(train_units) | set(val_units) == set(all_train_units)

print("Training engines:", len(train_units), train_units)
print("Validation engines:", len(val_units), val_units)

Retained sensors: ['s_2', 's_3', 's_4', 's_7', 's_8', 's_9', 's_11',
's_12', 's_13', 's_14', 's_15', 's_17', 's_20', 's_21']
Operating conditions fitted from all training engines: 4
Training engines: 80 [ 2  3  4  6  7  8  9 10 12 14 15 16
17 18 20 21 22 24
25 26 27 28 29 30 33 35 36 37 38 39 41 42 43 44 47
48
49 50 51 52 53 55 56 57 58 59 60 61 62 63 64 65 66
67
68 69 70 72 73 75 76 79 80 82 83 85 86 87 88 89 90
92
93 94 95 96 97 98 99 100]

```


Validation engines: 20 [1 5 11 13 19 23 31 32 34 40 45 46 54 71 74
77 78 81 84 91]

Generate train, validation, and final-test sequences

```
def gen_train_data(
    df: pd.DataFrame,
    sequence_length: int,
    columns: list[str],
):
    data = df[columns].to_numpy(dtype=np.float32)
    for stop in range(sequence_length, len(data) + 1):
        yield data[stop - sequence_length:stop]

def gen_data_wrapper(
    df: pd.DataFrame,
    sequence_length: int,
    columns: list[str],
    unit_nrs: np.ndarray,
) -> np.ndarray:
    sequences = [
        sequence
        for unit_nr in unit_nrs
        for sequence in gen_train_data(
            df[df["unit_nr"] == unit_nr],
            sequence_length,
            columns,
        )
    ]
    if not sequences:
        raise ValueError("No sequences were generated. Check sequence
length and dataset files.")
    return np.asarray(sequences, dtype=np.float32)

def gen_label_wrapper(
    df: pd.DataFrame,
    sequence_length: int,
    label: str,
    unit_nrs: np.ndarray,
) -> np.ndarray:
    labels = []
    for unit_nr in unit_nrs:
        unit_values = df.loc[df["unit_nr"] == unit_nr,
label].to_numpy(dtype=np.float32)
        labels.extend(unit_values[sequence_length - 1:])
    return np.asarray(labels, dtype=np.float32).reshape(-1, 1)
```

```

def gen_test_data(
    df: pd.DataFrame,
    sequence_length: int,
    columns: list[str],
    mask_value: float,
) -> np.ndarray:
    values = df[columns].to_numpy(dtype=np.float32)
    if len(values) < sequence_length:
        padded = np.full((sequence_length, len(columns)), mask_value,
dtype=np.float32)
        padded[-len(values):] = values
        values = padded
    return values[-sequence_length:]

train_split_array = gen_data_wrapper(
    train_scaled, SEQUENCE_LENGTH, remaining_sensors, train_units
)
train_split_label = gen_label_wrapper(
    train_scaled, SEQUENCE_LENGTH, "RUL", train_units
)
val_split_array = gen_data_wrapper(
    train_scaled, SEQUENCE_LENGTH, remaining_sensors, val_units
)
val_split_label = gen_label_wrapper(
    train_scaled, SEQUENCE_LENGTH, "RUL", val_units
)

test_units = np.sort(test_scaled["unit_nr"].unique())
test_array = np.asarray(
    [
        gen_test_data(
            test_scaled[test_scaled["unit_nr"] == unit_nr],
            SEQUENCE_LENGTH,
            remaining_sensors,
            MASK_VALUE,
        )
        for unit_nr in test_units
    ],
    dtype=np.float32,
)
test_label = y_test["RUL"].to_numpy(dtype=np.float32).reshape(-1, 1)

assert len(test_units) == len(test_label)
assert train_split_array.shape[0] == train_split_label.shape[0]
assert val_split_array.shape[0] == val_split_label.shape[0]
assert train_split_array.shape[1:] == (SEQUENCE_LENGTH,
len(remaining_sensors))
assert val_split_array.shape[1:] == (SEQUENCE_LENGTH,
len(remaining_sensors))

```

```

assert test_array.shape[1:] == (SEQUENCE_LENGTH,
len(remaining_sensors))

print("Training sequences:", train_split_array.shape,
train_split_label.shape)
print("Validation sequences:", val_split_array.shape,
val_split_label.shape)
print("Test sequences:", test_array.shape, test_label.shape)

Training sequences: (17692, 30, 14) (17692, 1)
Validation sequences: (4128, 30, 14) (4128, 1)
Test sequences: (100, 30, 14) (100, 1)

```

Build a test-like validation set

The exhaustive sliding-window validation set above takes every window along each validation engine's full trajectory. Because engines spend most of their life in the flat, capped-RUL region, this set is dominated by easy windows: 42% of its labels sit at the RUL cap of 125, versus only 15% for the official NASA test set (whose trajectories are truncated at an arbitrary cycle before failure, so they skew toward more-degraded states — test median RUL is 77.5 vs 108 for the exhaustive validation windows). Monitoring early stopping and checkpointing against the exhaustive set therefore selects for a distribution that is easier than what the model is ultimately scored on.

This cell builds a second validation set that mirrors the test set's construction: a handful of randomly-truncated windows per validation engine, using a fixed RNG seed independent of the model's training seed so every model (baseline and every ensemble member) is judged against the identical set of windows.

```

def gen_test_like_validation(
    df: pd.DataFrame,
    sequence_length: int,
    columns: list[str],
    label: str,
    unit_nrs: np.ndarray,
    rng: np.random.Generator,
    truncations_per_unit: int = 5,
) -> tuple[np.ndarray, np.ndarray]:
    """One or more randomly-truncated windows per engine, mirroring
    how the
    official test trajectories are cut off at an arbitrary cycle
    before
    failure, instead of the exhaustive every-window sliding approach.
    """
    arrays = []
    labels = []
    for unit_nr in unit_nrs:
        unit_df = df.loc[df["unit_nr"] == unit_nr, columns +
[label]].reset_index(drop=True)

```

```

        n_cycles = len(unit_df)
        if n_cycles < sequence_length:
            continue
        cutoffs = rng.integers(sequence_length, n_cycles + 1,
size=truncations_per_unit)
        for cutoff in cutoffs:
            window = unit_df.loc[cutoff - sequence_length:cutoff - 1,
columns].to_numpy(dtype=np.float32)
            arrays.append(window)
            labels.append(unit_df.loc[cutoff - 1, label])
    return (
        np.asarray(arrays, dtype=np.float32),
        np.asarray(labels, dtype=np.float32).reshape(-1, 1),
    )

```

```

TEST_LIKE_VAL_SEED = 12345

```

```

val_test_like_array, val_test_like_label = gen_test_like_validation(
    train_scaled,
    SEQUENCE_LENGTH,
    remaining_sensors,
    "RUL",
    val_units,
    np.random.default_rng(TEST_LIKE_VAL_SEED),
)

```

```

print("Test-like validation sequences:", val_test_like_array.shape,
val_test_like_label.shape)
print(
    "Test-like validation RUL: mean", val_test_like_label.mean(),
    "median", np.median(val_test_like_label),
    "frac at cap:", float((val_test_like_label >= RUL_CAP).mean()),
)
print(
    "Official test RUL: mean", test_label.mean(),
    "median", np.median(test_label),
    "frac at cap:", float((test_label >= RUL_CAP).mean()),
)

```

```

Test-like validation sequences: (100, 30, 14) (100, 1)
Test-like validation RUL: mean 85.56 median 97.0 frac at cap: 0.29
Official test RUL: mean 75.32 median 77.5 frac at cap: 0.15

```

Metrics

```

def regression_metrics(y_true: np.ndarray, y_pred: np.ndarray) ->
dict[str, float]:
    y_true = np.asarray(y_true).reshape(-1)
    y_pred = np.asarray(y_pred).reshape(-1)

```

```

    return {
        "rmse": float(np.sqrt(mean_squared_error(y_true, y_pred))),
        "mae": float(mean_absolute_error(y_true, y_pred)),
        "r2": float(r2_score(y_true, y_pred)),
    }

def print_metrics(label: str, metrics: dict[str, float]) -> None:
    print(
        f"{label}: RMSE={metrics['rmse']:.4f}, "
        f"MAE={metrics['mae']:.4f}, R2={metrics['r2']:.4f}"
    )

```

Build the parallel CNN–LSTM model

```

def zero_masked_timesteps(inputs: tf.Tensor) -> tf.Tensor:
    """Zero out padded timesteps before they reach a layer that
    ignores Keras masks.

    Conv1D does not propagate the mask produced by a Masking layer, so
    padded
    test trajectories (shorter than SEQUENCE_LENGTH) would otherwise
    feed the
    raw MASK_VALUE sentinel straight into the convolution. (For FD003
    the
    shortest official test trajectory is 38 cycles, longer than any
    SEQUENCE_LENGTH tried here, so this is currently a no-op in
    practice –
    kept because it is still the correct behavior if that ever
    changes.)
    """
    is_real_step = tf.reduce_any(tf.not_equal(inputs, MASK_VALUE),
    axis=-1, keepdims=True)
    return inputs * tf.cast(is_real_step, inputs.dtype)

def build_parallel_model(
    input_shape: tuple[int, int],
    dropout_rate: float = 0.2,
    l2_strength: float = 0.01,
) -> Model:
    """Parallel CNN-LSTM architecture reconstructed from the FD003
    experiment family.

    dropout_rate=0.2 and no inter-LSTM dropout, matching the original
    reconstruction: a run with dropout_rate=0.3 plus an inter-LSTM
    dropout
    layer was tried and measured worse (single-model test RMSE 15.95
    vs
    11.51 train, versus 15.10 test / 9.54 train here), so the extra

```

```

    regularization was reverted rather than kept on the strength of
theory
alone.
"""
input_cnn = Input(shape=input_shape, name="cnn_input")
input_lstm = Input(shape=input_shape, name="lstm_input")

# CNN branch: local degradation signatures.
# Conv1D can't consume a Keras mask, so padded steps are zeroed
explicitly
# instead of relying on a Masking layer (which would be a silent
no-op here).
cnn_branch = Lambda(zero_masked_timesteps, name="cnn_zero_mask")
(input_cnn)
cnn_branch = Conv1D(
    filters=128,
    kernel_size=3,
    padding="same",
    activation="relu",
    name="cnn_conv_1",
)(cnn_branch)
cnn_branch = MaxPooling1D(pool_size=2, padding="same",
name="cnn_pool")(cnn_branch)
cnn_branch = Flatten(name="cnn_flatten")(cnn_branch)
cnn_branch = Dense(
    128,
    activation="relu",
    kernel_regularizer=regularizers.l2(l2_strength),
    name="cnn_dense",
)(cnn_branch)
cnn_branch = Dropout(dropout_rate, name="cnn_dropout")(cnn_branch)

# LSTM branch: longer temporal degradation patterns.
lstm_branch = Masking(mask_value=MASK_VALUE, name="lstm_masking")
(input_lstm)
lstm_branch = LSTM(128, return_sequences=True, name="lstm_1")
(lstm_branch)
lstm_branch = LSTM(128, activation="tanh", name="lstm_2")
(lstm_branch)
lstm_branch = Dense(
    128,
    activation="relu",
    kernel_regularizer=regularizers.l2(l2_strength),
    name="lstm_dense",
)(lstm_branch)
lstm_branch = Dropout(dropout_rate, name="lstm_dropout")
(lstm_branch)

merged = Concatenate(name="feature_fusion")([cnn_branch,
lstm_branch])

```

```

merged = Dense(
    256,
    activation="relu",
    kernel_regularizer=regularizers.l2(l2_strength),
    name="merged_dense",
)(merged)
merged = Dropout(dropout_rate, name="merged_dropout")(merged)
output = Dense(1, activation="linear", name="rul_output")(merged)

model = Model([input_cnn, input_lstm], output,
name="fd003_parallel_cnn_lstm")
model.compile(optimizer=Adam(learning_rate=1e-3),
loss="mean_squared_error")
return model

```

```

parallel_model = build_parallel_model(
    (SEQUENCE_LENGTH, len(remaining_sensors))
)
parallel_model.summary()

```

Model: "fd003_parallel_cnn_lstm"

Layer (type)	Output Shape	Param #
Connected to		
=====	=====	=====
cnn_input (InputLayer)	[(None, 30, 14)]	0 []
cnn_zero_mask (Lambda)	(None, 30, 14)	0
['cnn_input[0][0]']		
lstm_input (InputLayer)	[(None, 30, 14)]	0 []
cnn_conv_1 (Conv1D)	(None, 30, 128)	5504
['cnn_zero_mask[0][0]']		
lstm_masking (Masking)	(None, 30, 14)	0
['lstm_input[0][0]']		
cnn_pool (MaxPooling1D)	(None, 15, 128)	0
['cnn_conv_1[0][0]']		

lstm_1 (LSTM)	(None, 30, 128)	73216
['lstm_masking[0][0]']		
cnn_flatten (Flatten)	(None, 1920)	0
['cnn_pool[0][0]']		
lstm_2 (LSTM)	(None, 128)	131584
['lstm_1[0][0]']		
cnn_dense (Dense)	(None, 128)	245888
['cnn_flatten[0][0]']		
lstm_dense (Dense)	(None, 128)	16512
['lstm_2[0][0]']		
cnn_dropout (Dropout)	(None, 128)	0
['cnn_dense[0][0]']		
lstm_dropout (Dropout)	(None, 128)	0
['lstm_dense[0][0]']		
feature_fusion (Concatenate)	(None, 256)	0
['cnn_dropout[0][0]',		
'lstm_dropout[0][0]']		
merged_dense (Dense)	(None, 256)	65792
['feature_fusion[0][0]']		
merged_dropout (Dropout)	(None, 256)	0
['merged_dense[0][0]']		
rul_output (Dense)	(None, 1)	257
['merged_dropout[0][0]']		

=====

=====


```
Total params: 538,753
Trainable params: 538,753
Non-trainable params: 0
```

Train with validation-based early stopping

```
PARALLEL_ARTIFACT_DIR = ARTIFACT_DIR / "parallel_reconstruction"
PARALLEL_ARTIFACT_DIR.mkdir(parents=True, exist_ok=True)
```

```
callbacks = [
    EarlyStopping(
        monitor="val_loss",
        mode="min",
        patience=15,
        restore_best_weights=True,
        verbose=1,
    ),
    ModelCheckpoint(
        PARALLEL_ARTIFACT_DIR / "best_parallel_model.keras",
        monitor="val_loss",
        mode="min",
        save_best_only=True,
        verbose=1,
    ),
    ReduceLROnPlateau(
        monitor="val_loss",
        factor=0.5,
        patience=5,
        min_lr=1e-5,
        verbose=1,
    ),
]
```

```
# Monitored on the exhaustive sliding-window validation set. A test-
like,
# randomly-truncated validation set was tried for monitoring instead
(on the
# theory that it better matches how the official test trajectories are
# truncated) and measured worse in practice (single-model test RMSE
16.39
# vs 15.10 here) - the smaller, truncation-based set was too noisy a
signal
# for early stopping/checkpoint selection, so monitoring was reverted
to
# this larger set. val_test_like_array/label are still evaluated below
for
# comparison, just not used to drive training decisions.
```

```
history = parallel_model.fit(  
    [train_split_array, train_split_array],  
    train_split_label,  
    validation_data=(val_split_array, val_split_array),  
    val_split_label),  
    epochs=50,  
    batch_size=32,  
    callbacks=callbacks,  
    verbose=2,  
)
```

Epoch 1/50

Epoch 1: val_loss improved from inf to 220.55519, saving model to C:\Users\Kamal\Documents\AUC\thesiswork\env_eweda\predictive-maintenance-rul_working\artifacts\cmapss_fd003\parallel_reconstruction\best_parallel_model.keras

553/553 - 31s - loss: 572.2037 - val_loss: 220.5552 - lr: 0.0010 - 31s/epoch - 55ms/step

Epoch 2/50

Epoch 2: val_loss improved from 220.55519 to 163.65286, saving model to C:\Users\Kamal\Documents\AUC\thesiswork\env_eweda\predictive-maintenance-rul_working\artifacts\cmapss_fd003\parallel_reconstruction\best_parallel_model.keras

553/553 - 17s - loss: 203.9356 - val_loss: 163.6529 - lr: 0.0010 - 17s/epoch - 31ms/step

Epoch 3/50

Epoch 3: val_loss improved from 163.65286 to 162.59624, saving model to C:\Users\Kamal\Documents\AUC\thesiswork\env_eweda\predictive-maintenance-rul_working\artifacts\cmapss_fd003\parallel_reconstruction\best_parallel_model.keras

553/553 - 16s - loss: 177.0548 - val_loss: 162.5962 - lr: 0.0010 - 16s/epoch - 29ms/step

Epoch 4/50

Epoch 4: val_loss improved from 162.59624 to 150.96045, saving model to C:\Users\Kamal\Documents\AUC\thesiswork\env_eweda\predictive-maintenance-rul_working\artifacts\cmapss_fd003\parallel_reconstruction\best_parallel_model.keras

553/553 - 15s - loss: 162.5770 - val_loss: 150.9604 - lr: 0.0010 - 15s/epoch - 27ms/step

Epoch 5/50

Epoch 5: val_loss did not improve from 150.96045

553/553 - 17s - loss: 153.3459 - val_loss: 182.1761 - lr: 0.0010 - 17s/epoch - 32ms/step

Epoch 6/50

Epoch 6: val_loss did not improve from 150.96045
553/553 - 16s - loss: 147.6423 - val_loss: 170.6849 - lr: 0.0010 -
16s/epoch - 29ms/step
Epoch 7/50

Epoch 7: val_loss did not improve from 150.96045
553/553 - 16s - loss: 132.5159 - val_loss: 175.7621 - lr: 0.0010 -
16s/epoch - 29ms/step
Epoch 8/50

Epoch 8: val_loss did not improve from 150.96045
553/553 - 16s - loss: 131.0446 - val_loss: 156.2091 - lr: 0.0010 -
16s/epoch - 30ms/step
Epoch 9/50

Epoch 9: val_loss did not improve from 150.96045

Epoch 9: ReduceLROnPlateau reducing learning rate to
0.0005000000237487257.
553/553 - 17s - loss: 120.7654 - val_loss: 166.7859 - lr: 0.0010 -
17s/epoch - 30ms/step
Epoch 10/50

Epoch 10: val_loss did not improve from 150.96045
553/553 - 16s - loss: 95.2550 - val_loss: 173.8885 - lr: 5.0000e-04 -
16s/epoch - 30ms/step
Epoch 11/50

Epoch 11: val_loss did not improve from 150.96045
553/553 - 17s - loss: 89.5241 - val_loss: 176.2779 - lr: 5.0000e-04 -
17s/epoch - 31ms/step
Epoch 12/50

Epoch 12: val_loss did not improve from 150.96045
553/553 - 16s - loss: 80.9632 - val_loss: 183.5932 - lr: 5.0000e-04 -
16s/epoch - 29ms/step
Epoch 13/50

Epoch 13: val_loss did not improve from 150.96045
553/553 - 19s - loss: 79.4113 - val_loss: 182.5824 - lr: 5.0000e-04 -
19s/epoch - 35ms/step
Epoch 14/50

Epoch 14: val_loss did not improve from 150.96045

Epoch 14: ReduceLROnPlateau reducing learning rate to
0.0002500000118743628.
553/553 - 18s - loss: 74.9614 - val_loss: 178.6133 - lr: 5.0000e-04 -
18s/epoch - 33ms/step
Epoch 15/50

```
Epoch 15: val_loss did not improve from 150.96045
553/553 - 16s - loss: 64.3863 - val_loss: 184.3649 - lr: 2.5000e-04 -
16s/epoch - 29ms/step
Epoch 16/50

Epoch 16: val_loss did not improve from 150.96045
553/553 - 17s - loss: 61.2063 - val_loss: 183.6932 - lr: 2.5000e-04 -
17s/epoch - 31ms/step
Epoch 17/50

Epoch 17: val_loss did not improve from 150.96045
553/553 - 15s - loss: 59.3287 - val_loss: 186.5795 - lr: 2.5000e-04 -
15s/epoch - 28ms/step
Epoch 18/50

Epoch 18: val_loss did not improve from 150.96045
553/553 - 16s - loss: 57.9485 - val_loss: 189.7922 - lr: 2.5000e-04 -
16s/epoch - 28ms/step
Epoch 19/50
Restoring model weights from the end of the best epoch: 4.

Epoch 19: val_loss did not improve from 150.96045

Epoch 19: ReduceLROnPlateau reducing learning rate to
0.0001250000059371814.
553/553 - 15s - loss: 56.5271 - val_loss: 199.9091 - lr: 2.5000e-04 -
15s/epoch - 27ms/step
Epoch 19: early stopping
```

Evaluate the selected model

```
train_prediction = parallel_model.predict(
    [train_split_array, train_split_array], verbose=0
)
val_prediction = parallel_model.predict(
    [val_split_array, val_split_array], verbose=0
)
val_test_like_prediction = parallel_model.predict(
    [val_test_like_array, val_test_like_array], verbose=0
)

# The official test set is touched only here, after architecture and
# stopping decisions.
test_prediction = parallel_model.predict([test_array, test_array],
    verbose=0)

metrics = {
    "train": regression_metrics(train_split_label, train_prediction),
    "validation_exhaustive_windows":
```

```

regression_metrics(val_split_label, val_prediction),
    "validation_test_like": regression_metrics(val_test_like_label,
val_test_like_prediction),
    "test": regression_metrics(test_label, test_prediction),
}

for split_name, split_metrics in metrics.items():
    print_metrics(split_name, split_metrics)

results = {
    "status": "clean rerun",
    "dataset": "NASA C-MAPSS FD003",
    "model": "parallel CNN-LSTM reconstruction",
    "thesis_reported_test_rmse_reference": 14.75,
    "seed": SEED,
    "sequence_length": SEQUENCE_LENGTH,
    "rul_cap": RUL_CAP,
    "train_units": train_units.tolist(),
    "validation_units": val_units.tolist(),
    "metrics": metrics,
}

with (PARALLEL_ARTIFACT_DIR / "metrics.json").open("w", encoding="utf-
8") as handle:
    json.dump(results, handle, indent=2)

print("Saved metrics to", PARALLEL_ARTIFACT_DIR / "metrics.json")

train: RMSE=10.0729, MAE=7.1284, R2=0.9407
validation_exhaustive_windows: RMSE=12.0770, MAE=8.7834, R2=0.9160
validation_test_like: RMSE=12.2380, MAE=9.1435, R2=0.9009
test: RMSE=14.6804, MAE=11.0421, R2=0.8742
Saved metrics to C:\Users\Kamal\Documents\AUC\thesiswork\env_eweda\
predictive-maintenance-rul_working\artifacts\cmapss_fd003\
parallel_reconstruction\metrics.json

```

Training history and RUL predictions

```

fig, axes = plt.subplots(1, 2, figsize=(16, 5))

axes[0].plot(history.history["loss"], label="Training loss")
axes[0].plot(history.history["val_loss"], label="Validation loss")
axes[0].set_title("Training and validation loss")
axes[0].set_xlabel("Epoch")
axes[0].set_ylabel("MSE loss")
axes[0].legend()

axes[1].plot(test_units, test_label.reshape(-1), label="Actual RUL",
linewidth=2)
axes[1].plot(test_units, test_prediction.reshape(-1), label="Predicted

```

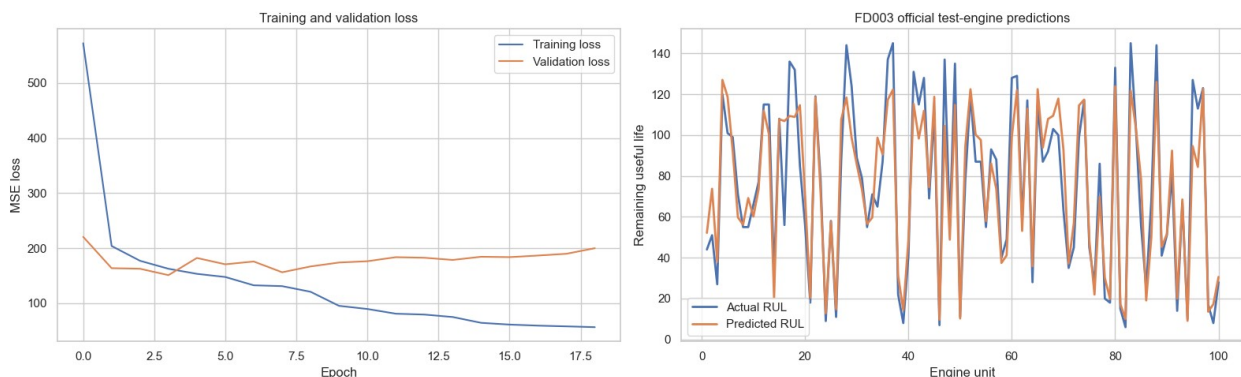
```

RUL", linewidth=2)
axes[1].set_title("FD003 official test-engine predictions")
axes[1].set_xlabel("Engine unit")
axes[1].set_ylabel("Remaining useful life")
axes[1].legend()

fig.tight_layout()
figure_path = PARALLEL_ARTIFACT_DIR /
"parallel_training_and_test_predictions.png"
fig.savefig(figure_path, dpi=180, bbox_inches="tight")
plt.show()

print("Saved figure to", figure_path)

```



Saved figure to C:\Users\Kamal\Documents\AUC\thesiswork\env_eweda\predictive-maintenance-rul_working\artifacts\cmapss_fd003\parallel_reconstruction\parallel_training_and_test_predictions.png

Seed ensemble

Trains several independently-initialized copies of the same architecture and averages their test predictions. This cancels initialization/optimization noise rather than fixing any single systematic issue, so it stacks with the CNN masking fix and the higher dropout above. Each member is a full 50-epoch run with the same early-stopping setup as the single-model baseline, so this cell takes roughly `len(ENSEMBLE_SEEDS)` times as long as the baseline training cell above.

```

ENSEMBLE_SEEDS = [42, 7, 123, 2024, 31415]

ensemble_train_predictions = []
ensemble_val_predictions = []
ensemble_val_test_like_predictions = []
ensemble_test_predictions = []
ensemble_histories = {}

for member_seed in ENSEMBLE_SEEDS:
    print(f"\n=== Training ensemble member, seed={member_seed} ===")

```

```

    set_all_seeds(member_seed)

    member_model = build_parallel_model((SEQUENCE_LENGTH,
len(remaining_sensors)))

    member_checkpoint_path = PARALLEL_ARTIFACT_DIR /
f"ensemble_member_seed{member_seed}.keras"
    member_callbacks = [
        EarlyStopping(
            monitor="val_loss",
            mode="min",
            patience=15,
            restore_best_weights=True,
            verbose=0,
        ),
        ModelCheckpoint(
            member_checkpoint_path,
            monitor="val_loss",
            mode="min",
            save_best_only=True,
            verbose=0,
        ),
        ReduceLROnPlateau(
            monitor="val_loss",
            factor=0.5,
            patience=5,
            min_lr=1e-5,
            verbose=0,
        ),
    ]

    # Monitored on the exhaustive validation set, same as the single-
    # model baseline.
    member_history = member_model.fit(
        [train_split_array, train_split_array],
        train_split_label,
        validation_data=([val_split_array, val_split_array],
val_split_label),
        epochs=50,
        batch_size=32,
        callbacks=member_callbacks,
        verbose=2,
    )
    ensemble_histories[member_seed] = member_history.history

    ensemble_train_predictions.append(
        member_model.predict([train_split_array, train_split_array],
verbose=0)
    )
    ensemble_val_predictions.append(

```

```

        member_model.predict([val_split_array, val_split_array],
verbose=0)
    )
    ensemble_val_test_like_predictions.append(
        member_model.predict([val_test_like_array,
val_test_like_array], verbose=0)
    )
    member_test_prediction = member_model.predict([test_array,
test_array], verbose=0)
    ensemble_test_predictions.append(member_test_prediction)

    print_metrics(f"seed {member_seed} test",
regression_metrics(test_label, member_test_prediction))

```

=== Training ensemble member, seed=42 ===

Epoch 1/50

553/553 - 28s - loss: 572.2037 - val_loss: 220.5552 - lr: 0.0010 -
28s/epoch - 51ms/step

Epoch 2/50

553/553 - 15s - loss: 203.9356 - val_loss: 163.6529 - lr: 0.0010 -
15s/epoch - 27ms/step

Epoch 3/50

553/553 - 16s - loss: 177.0548 - val_loss: 162.5962 - lr: 0.0010 -
16s/epoch - 28ms/step

Epoch 4/50

553/553 - 15s - loss: 162.5770 - val_loss: 150.9604 - lr: 0.0010 -
15s/epoch - 27ms/step

Epoch 5/50

553/553 - 16s - loss: 153.3459 - val_loss: 182.1761 - lr: 0.0010 -
16s/epoch - 28ms/step

Epoch 6/50

553/553 - 17s - loss: 147.6423 - val_loss: 170.6849 - lr: 0.0010 -
17s/epoch - 30ms/step

Epoch 7/50

553/553 - 16s - loss: 132.5159 - val_loss: 175.7621 - lr: 0.0010 -
16s/epoch - 30ms/step

Epoch 8/50

553/553 - 17s - loss: 131.0446 - val_loss: 156.2091 - lr: 0.0010 -
17s/epoch - 31ms/step

Epoch 9/50

553/553 - 16s - loss: 120.7654 - val_loss: 166.7859 - lr: 0.0010 -
16s/epoch - 29ms/step

Epoch 10/50

553/553 - 17s - loss: 95.2550 - val_loss: 173.8885 - lr: 5.0000e-04 -
17s/epoch - 31ms/step

Epoch 11/50

553/553 - 16s - loss: 89.5241 - val_loss: 176.2779 - lr: 5.0000e-04 -
16s/epoch - 29ms/step

Epoch 12/50

553/553 - 18s - loss: 80.9632 - val_loss: 183.5932 - lr: 5.0000e-04 -
18s/epoch - 33ms/step
Epoch 13/50
553/553 - 17s - loss: 79.4113 - val_loss: 182.5824 - lr: 5.0000e-04 -
17s/epoch - 31ms/step
Epoch 14/50
553/553 - 21s - loss: 74.9614 - val_loss: 178.6133 - lr: 5.0000e-04 -
21s/epoch - 38ms/step
Epoch 15/50
553/553 - 20s - loss: 64.3863 - val_loss: 184.3649 - lr: 2.5000e-04 -
20s/epoch - 36ms/step
Epoch 16/50
553/553 - 17s - loss: 61.2063 - val_loss: 183.6932 - lr: 2.5000e-04 -
17s/epoch - 30ms/step
Epoch 17/50
553/553 - 18s - loss: 59.3287 - val_loss: 186.5795 - lr: 2.5000e-04 -
18s/epoch - 33ms/step
Epoch 18/50
553/553 - 22s - loss: 57.9485 - val_loss: 189.7922 - lr: 2.5000e-04 -
22s/epoch - 40ms/step
Epoch 19/50
553/553 - 21s - loss: 56.5271 - val_loss: 199.9091 - lr: 2.5000e-04 -
21s/epoch - 38ms/step
seed 42 test: RMSE=14.6804, MAE=11.0421, R2=0.8742

=== Training ensemble member, seed=7 ===

Epoch 1/50
553/553 - 30s - loss: 639.1831 - val_loss: 187.9745 - lr: 0.0010 -
30s/epoch - 53ms/step
Epoch 2/50
553/553 - 17s - loss: 206.9793 - val_loss: 172.1889 - lr: 0.0010 -
17s/epoch - 30ms/step
Epoch 3/50
553/553 - 17s - loss: 192.7467 - val_loss: 183.0970 - lr: 0.0010 -
17s/epoch - 31ms/step
Epoch 4/50
553/553 - 16s - loss: 174.9696 - val_loss: 152.6492 - lr: 0.0010 -
16s/epoch - 29ms/step
Epoch 5/50
553/553 - 17s - loss: 168.0298 - val_loss: 153.6228 - lr: 0.0010 -
17s/epoch - 30ms/step
Epoch 6/50
553/553 - 17s - loss: 166.2405 - val_loss: 144.0235 - lr: 0.0010 -
17s/epoch - 31ms/step
Epoch 7/50
553/553 - 17s - loss: 160.5514 - val_loss: 151.6686 - lr: 0.0010 -
17s/epoch - 30ms/step
Epoch 8/50
553/553 - 18s - loss: 152.3132 - val_loss: 159.3912 - lr: 0.0010 -

18s/epoch - 32ms/step
Epoch 9/50
553/553 - 16s - loss: 146.2944 - val_loss: 179.3222 - lr: 0.0010 -
16s/epoch - 29ms/step
Epoch 10/50
553/553 - 19s - loss: 144.3987 - val_loss: 187.9096 - lr: 0.0010 -
19s/epoch - 34ms/step
Epoch 11/50
553/553 - 16s - loss: 144.6025 - val_loss: 154.0117 - lr: 0.0010 -
16s/epoch - 29ms/step
Epoch 12/50
553/553 - 17s - loss: 117.1998 - val_loss: 157.2592 - lr: 5.0000e-04 -
17s/epoch - 31ms/step
Epoch 13/50
553/553 - 16s - loss: 110.6386 - val_loss: 156.0686 - lr: 5.0000e-04 -
16s/epoch - 29ms/step
Epoch 14/50
553/553 - 18s - loss: 108.7939 - val_loss: 157.2064 - lr: 5.0000e-04 -
18s/epoch - 33ms/step
Epoch 15/50
553/553 - 18s - loss: 104.5088 - val_loss: 170.7978 - lr: 5.0000e-04 -
18s/epoch - 33ms/step
Epoch 16/50
553/553 - 16s - loss: 103.0635 - val_loss: 161.3326 - lr: 5.0000e-04 -
16s/epoch - 29ms/step
Epoch 17/50
553/553 - 18s - loss: 92.5687 - val_loss: 168.4046 - lr: 2.5000e-04 -
18s/epoch - 32ms/step
Epoch 18/50
553/553 - 16s - loss: 87.7113 - val_loss: 157.7384 - lr: 2.5000e-04 -
16s/epoch - 30ms/step
Epoch 19/50
553/553 - 18s - loss: 84.8267 - val_loss: 157.5283 - lr: 2.5000e-04 -
18s/epoch - 33ms/step
Epoch 20/50
553/553 - 17s - loss: 84.0134 - val_loss: 167.0105 - lr: 2.5000e-04 -
17s/epoch - 30ms/step
Epoch 21/50
553/553 - 18s - loss: 82.6175 - val_loss: 159.5110 - lr: 2.5000e-04 -
18s/epoch - 33ms/step
seed 7 test: RMSE=14.3526, MAE=10.3687, R2=0.8798

=== Training ensemble member, seed=123 ===

Epoch 1/50
553/553 - 29s - loss: 620.3268 - val_loss: 190.8225 - lr: 0.0010 -
29s/epoch - 53ms/step
Epoch 2/50
553/553 - 17s - loss: 213.9067 - val_loss: 183.6433 - lr: 0.0010 -
17s/epoch - 31ms/step

Epoch 3/50
553/553 - 18s - loss: 181.2715 - val_loss: 169.4840 - lr: 0.0010 -
18s/epoch - 33ms/step

Epoch 4/50
553/553 - 17s - loss: 172.2265 - val_loss: 176.1844 - lr: 0.0010 -
17s/epoch - 31ms/step

Epoch 5/50
553/553 - 18s - loss: 165.9416 - val_loss: 159.4808 - lr: 0.0010 -
18s/epoch - 33ms/step

Epoch 6/50
553/553 - 18s - loss: 156.0704 - val_loss: 149.2225 - lr: 0.0010 -
18s/epoch - 33ms/step

Epoch 7/50
553/553 - 26s - loss: 154.0030 - val_loss: 148.1037 - lr: 0.0010 -
26s/epoch - 46ms/step

Epoch 8/50
553/553 - 18s - loss: 144.1512 - val_loss: 156.0125 - lr: 0.0010 -
18s/epoch - 33ms/step

Epoch 9/50
553/553 - 17s - loss: 138.7018 - val_loss: 170.0659 - lr: 0.0010 -
17s/epoch - 30ms/step

Epoch 10/50
553/553 - 20s - loss: 137.8036 - val_loss: 179.1326 - lr: 0.0010 -
20s/epoch - 36ms/step

Epoch 11/50
553/553 - 22s - loss: 133.1301 - val_loss: 177.6828 - lr: 0.0010 -
22s/epoch - 39ms/step

Epoch 12/50
553/553 - 18s - loss: 128.4164 - val_loss: 170.8297 - lr: 0.0010 -
18s/epoch - 32ms/step

Epoch 13/50
553/553 - 29s - loss: 108.4640 - val_loss: 172.0673 - lr: 5.0000e-04 -
29s/epoch - 52ms/step

Epoch 14/50
553/553 - 22s - loss: 100.5664 - val_loss: 174.9181 - lr: 5.0000e-04 -
22s/epoch - 40ms/step

Epoch 15/50
553/553 - 22s - loss: 100.1471 - val_loss: 179.7613 - lr: 5.0000e-04 -
22s/epoch - 39ms/step

Epoch 16/50
553/553 - 25s - loss: 98.1001 - val_loss: 175.0053 - lr: 5.0000e-04 -
25s/epoch - 45ms/step

Epoch 17/50
553/553 - 20s - loss: 95.7308 - val_loss: 176.7522 - lr: 5.0000e-04 -
20s/epoch - 35ms/step

Epoch 18/50
553/553 - 27s - loss: 85.0903 - val_loss: 180.0822 - lr: 2.5000e-04 -
27s/epoch - 49ms/step

Epoch 19/50

553/553 - 24s - loss: 82.7771 - val_loss: 185.7628 - lr: 2.5000e-04 -
24s/epoch - 44ms/step
Epoch 20/50
553/553 - 23s - loss: 81.7281 - val_loss: 175.8440 - lr: 2.5000e-04 -
23s/epoch - 41ms/step
Epoch 21/50
553/553 - 23s - loss: 78.5684 - val_loss: 179.1053 - lr: 2.5000e-04 -
23s/epoch - 41ms/step
Epoch 22/50
553/553 - 21s - loss: 79.2143 - val_loss: 174.6354 - lr: 2.5000e-04 -
21s/epoch - 39ms/step
seed 123 test: RMSE=13.9141, MAE=10.3571, R2=0.8870

=== Training ensemble member, seed=2024 ===

Epoch 1/50
553/553 - 35s - loss: 613.9796 - val_loss: 189.7876 - lr: 0.0010 -
35s/epoch - 63ms/step
Epoch 2/50
553/553 - 21s - loss: 211.4695 - val_loss: 164.1658 - lr: 0.0010 -
21s/epoch - 38ms/step
Epoch 3/50
553/553 - 21s - loss: 185.5949 - val_loss: 170.6150 - lr: 0.0010 -
21s/epoch - 38ms/step
Epoch 4/50
553/553 - 19s - loss: 173.3797 - val_loss: 158.0312 - lr: 0.0010 -
19s/epoch - 35ms/step
Epoch 5/50
553/553 - 18s - loss: 170.0522 - val_loss: 141.8553 - lr: 0.0010 -
18s/epoch - 33ms/step
Epoch 6/50
553/553 - 20s - loss: 154.6285 - val_loss: 167.0750 - lr: 0.0010 -
20s/epoch - 36ms/step
Epoch 7/50
553/553 - 19s - loss: 151.8862 - val_loss: 135.2002 - lr: 0.0010 -
19s/epoch - 34ms/step
Epoch 8/50
553/553 - 20s - loss: 145.7396 - val_loss: 140.8326 - lr: 0.0010 -
20s/epoch - 37ms/step
Epoch 9/50
553/553 - 18s - loss: 137.8065 - val_loss: 200.6170 - lr: 0.0010 -
18s/epoch - 33ms/step
Epoch 10/50
553/553 - 20s - loss: 129.0537 - val_loss: 174.1409 - lr: 0.0010 -
20s/epoch - 37ms/step
Epoch 11/50
553/553 - 18s - loss: 123.9337 - val_loss: 171.2782 - lr: 0.0010 -
18s/epoch - 32ms/step
Epoch 12/50
553/553 - 25s - loss: 122.8236 - val_loss: 190.6129 - lr: 0.0010 -

25s/epoch - 45ms/step
Epoch 13/50
553/553 - 22s - loss: 101.1089 - val_loss: 186.9343 - lr: 5.0000e-04 -
22s/epoch - 40ms/step
Epoch 14/50
553/553 - 19s - loss: 94.2956 - val_loss: 163.0250 - lr: 5.0000e-04 -
19s/epoch - 34ms/step
Epoch 15/50
553/553 - 17s - loss: 89.6841 - val_loss: 183.2348 - lr: 5.0000e-04 -
17s/epoch - 32ms/step
Epoch 16/50
553/553 - 18s - loss: 87.2423 - val_loss: 175.6244 - lr: 5.0000e-04 -
18s/epoch - 32ms/step
Epoch 17/50
553/553 - 16s - loss: 84.7677 - val_loss: 188.0899 - lr: 5.0000e-04 -
16s/epoch - 30ms/step
Epoch 18/50
553/553 - 17s - loss: 74.1592 - val_loss: 188.0116 - lr: 2.5000e-04 -
17s/epoch - 31ms/step
Epoch 19/50
553/553 - 16s - loss: 72.3686 - val_loss: 180.2269 - lr: 2.5000e-04 -
16s/epoch - 30ms/step
Epoch 20/50
553/553 - 17s - loss: 71.0661 - val_loss: 195.6462 - lr: 2.5000e-04 -
17s/epoch - 31ms/step
Epoch 21/50
553/553 - 18s - loss: 68.7437 - val_loss: 168.4549 - lr: 2.5000e-04 -
18s/epoch - 32ms/step
Epoch 22/50
553/553 - 16s - loss: 67.2147 - val_loss: 187.1555 - lr: 2.5000e-04 -
16s/epoch - 29ms/step
seed 2024 test: RMSE=13.8414, MAE=10.3188, R2=0.8882

=== Training ensemble member, seed=31415 ===

Epoch 1/50
553/553 - 29s - loss: 597.0151 - val_loss: 228.5856 - lr: 0.0010 -
29s/epoch - 52ms/step
Epoch 2/50
553/553 - 16s - loss: 235.8727 - val_loss: 205.6139 - lr: 0.0010 -
16s/epoch - 29ms/step
Epoch 3/50
553/553 - 17s - loss: 195.9462 - val_loss: 206.2327 - lr: 0.0010 -
17s/epoch - 30ms/step
Epoch 4/50
553/553 - 16s - loss: 178.4738 - val_loss: 165.2509 - lr: 0.0010 -
16s/epoch - 29ms/step
Epoch 5/50
553/553 - 18s - loss: 168.5851 - val_loss: 155.4637 - lr: 0.0010 -
18s/epoch - 33ms/step

Epoch 6/50
553/553 - 20s - loss: 166.4254 - val_loss: 155.0165 - lr: 0.0010 -
20s/epoch - 36ms/step
Epoch 7/50
553/553 - 17s - loss: 162.9892 - val_loss: 162.2055 - lr: 0.0010 -
17s/epoch - 30ms/step
Epoch 8/50
553/553 - 24s - loss: 147.1697 - val_loss: 171.1219 - lr: 0.0010 -
24s/epoch - 43ms/step
Epoch 9/50
553/553 - 18s - loss: 142.4947 - val_loss: 152.9318 - lr: 0.0010 -
18s/epoch - 33ms/step
Epoch 10/50
553/553 - 17s - loss: 132.3128 - val_loss: 146.1176 - lr: 0.0010 -
17s/epoch - 30ms/step
Epoch 11/50
553/553 - 17s - loss: 125.7193 - val_loss: 152.2254 - lr: 0.0010 -
17s/epoch - 31ms/step
Epoch 12/50
553/553 - 16s - loss: 119.7438 - val_loss: 178.6454 - lr: 0.0010 -
16s/epoch - 29ms/step
Epoch 13/50
553/553 - 18s - loss: 117.6096 - val_loss: 167.2109 - lr: 0.0010 -
18s/epoch - 32ms/step
Epoch 14/50
553/553 - 16s - loss: 111.8168 - val_loss: 196.3474 - lr: 0.0010 -
16s/epoch - 30ms/step
Epoch 15/50
553/553 - 17s - loss: 108.0188 - val_loss: 189.8116 - lr: 0.0010 -
17s/epoch - 31ms/step
Epoch 16/50
553/553 - 16s - loss: 97.1011 - val_loss: 188.5026 - lr: 5.0000e-04 -
16s/epoch - 29ms/step
Epoch 17/50
553/553 - 17s - loss: 90.6950 - val_loss: 215.8786 - lr: 5.0000e-04 -
17s/epoch - 31ms/step
Epoch 18/50
553/553 - 17s - loss: 91.5438 - val_loss: 198.5791 - lr: 5.0000e-04 -
17s/epoch - 30ms/step
Epoch 19/50
553/553 - 17s - loss: 87.7538 - val_loss: 215.5959 - lr: 5.0000e-04 -
17s/epoch - 31ms/step
Epoch 20/50
553/553 - 16s - loss: 83.4166 - val_loss: 196.7286 - lr: 5.0000e-04 -
16s/epoch - 29ms/step
Epoch 21/50
553/553 - 16s - loss: 75.0117 - val_loss: 214.9553 - lr: 2.5000e-04 -
16s/epoch - 28ms/step
Epoch 22/50

```
553/553 - 17s - loss: 71.7805 - val_loss: 196.2033 - lr: 2.5000e-04 -  
17s/epoch - 30ms/step  
Epoch 23/50  
553/553 - 16s - loss: 68.9525 - val_loss: 214.4947 - lr: 2.5000e-04 -  
16s/epoch - 28ms/step  
Epoch 24/50  
553/553 - 16s - loss: 69.3510 - val_loss: 204.4624 - lr: 2.5000e-04 -  
16s/epoch - 30ms/step  
Epoch 25/50  
553/553 - 16s - loss: 66.6600 - val_loss: 208.4608 - lr: 2.5000e-04 -  
16s/epoch - 28ms/step  
seed 31415 test: RMSE=14.1958, MAE=10.3107, R2=0.8824
```

```
ensemble_train_prediction = np.mean(ensemble_train_predictions,  
axis=0)  
ensemble_val_prediction = np.mean(ensemble_val_predictions, axis=0)  
ensemble_val_test_like_prediction =  
np.mean(ensemble_val_test_like_predictions, axis=0)  
ensemble_test_prediction = np.mean(ensemble_test_predictions, axis=0)  
  
ensemble_metrics = {  
    "train": regression_metrics(train_split_label,  
ensemble_train_prediction),  
    "validation_exhaustive_windows":  
regression_metrics(val_split_label, ensemble_val_prediction),  
    "validation_test_like": regression_metrics(val_test_like_label,  
ensemble_val_test_like_prediction),  
    "test": regression_metrics(test_label, ensemble_test_prediction),  
}  
  
for split_name, split_metrics in ensemble_metrics.items():  
    print_metrics(f"ensemble {split_name}", split_metrics)  
  
per_member_test_metrics = [  
    regression_metrics(test_label, member_prediction)  
    for member_prediction in ensemble_test_predictions  
]  
  
ensemble_results = {  
    "status": "clean rerun",  
    "dataset": "NASA C-MAPSS FD003",  
    "model": "parallel CNN-LSTM reconstruction (seed ensemble, mean of  
predictions)",  
    "ensemble_seeds": ENSEMBLE_SEEDS,  
    "sequence_length": SEQUENCE_LENGTH,  
    "rul_cap": RUL_CAP,  
    "train_units": train_units.tolist(),  
    "validation_units": val_units.tolist(),  
    "ensemble_metrics": ensemble_metrics,  
    "per_member_test_metrics": per_member_test_metrics,
```

```

}

with (PARALLEL_ARTIFACT_DIR / "ensemble_metrics.json").open("w",
encoding="utf-8") as handle:
    json.dump(ensemble_results, handle, indent=2)

print("Saved ensemble metrics to", PARALLEL_ARTIFACT_DIR /
"ensemble_metrics.json")

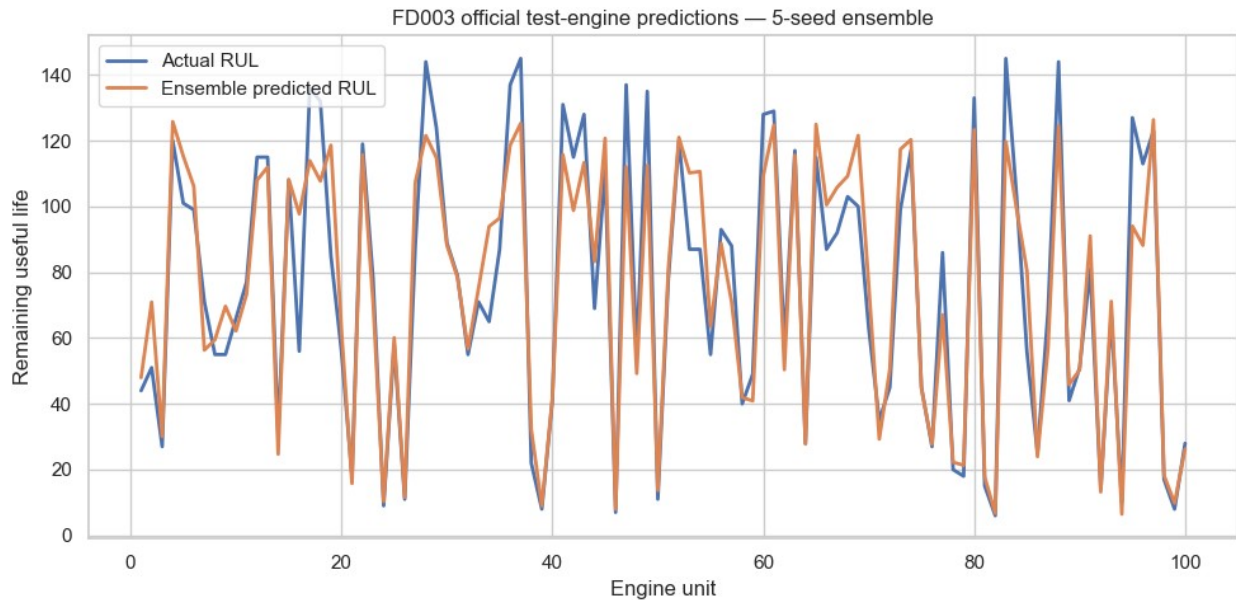
ensemble train: RMSE=8.6454, MAE=5.9984, R2=0.9563
ensemble validation_exhaustive_windows: RMSE=11.1007, MAE=7.7356,
R2=0.9290
ensemble validation_test_like: RMSE=11.4007, MAE=8.1708, R2=0.9140
ensemble test: RMSE=13.3910, MAE=9.7932, R2=0.8954
Saved ensemble metrics to C:\Users\Kamal\Documents\AUC\thesiswork\
env_eweda\predictive-maintenance-rul_working\artifacts\cmapss_fd003\
parallel_reconstruction\ensemble_metrics.json

fig, ax = plt.subplots(figsize=(10, 5))
ax.plot(test_units, test_label.reshape(-1), label="Actual RUL",
linewidth=2)
ax.plot(test_units, ensemble_test_prediction.reshape(-1),
label="Ensemble predicted RUL", linewidth=2)
ax.set_title(f"FD003 official test-engine predictions -
{len(ENSEMBLE_SEEDS)}-seed ensemble")
ax.set_xlabel("Engine unit")
ax.set_ylabel("Remaining useful life")
ax.legend()
fig.tight_layout()

ensemble_figure_path = PARALLEL_ARTIFACT_DIR /
"ensemble_test_predictions.png"
fig.savefig(ensemble_figure_path, dpi=180, bbox_inches="tight")
plt.show()

print("Saved figure to", ensemble_figure_path)

```

Saved figure to C:\Users\Kamal\Documents\AUC\thesiswork\env_eweda\predictive-maintenance-rul_working\artifacts\cmapss_fd003\parallel_reconstruction\ensemble_test_predictions.png

Capture the execution environment

```
environment = {
    "python": sys.version,
    "platform": platform.platform(),
    "tensorflow": tf.__version__,
    "numpy": np.__version__,
    "pandas": pd.__version__,
    "scikit_learn": sklearn.__version__,
    "seed": SEED,
}

with (ARTIFACT_DIR / "environment.json").open("w", encoding="utf-8")
as handle:
    json.dump(environment, handle, indent=2)

environment

{'python': '3.8.0 (tags/v3.8.0:fa919fd, Oct 14 2019, 19:37:50) [MSC
v.1916 64 bit (AMD64)]',
 'platform': 'Windows-10-10.0.26100-SP0',
 'tensorflow': '2.10.0',
 'numpy': '1.23.5',
 'pandas': '2.0.3',
 'scikit_learn': '1.3.2',
 'seed': 42}
```